Design of a Smart Wearable System for Gold Jewellery Hallmark Recognition and Readability Assessment

Wearable Computing (BCSE315L) course project

Priyanka Lingesan, Riddhi Solanki, Pranathi Kaparthi, Krishnaraj N

Vellore Institute of Technology, Vellore, India

SIMULATION ONLY: no physical prototype | SYNTHETIC hallmark-style inscriptions | reads characters; NOT a purity, authenticity or BIS-certification check

Introduction and Motivation

- Gold jewellery carries small hallmark-style inscriptions: engraved / stamped / embossed / laser-marked
- Hard to read by eye: tiny, low contrast, worn, glare
- Failure mode: confident but wrong OCR output; a silent misread costs more than a retake
- Wrist camera: hands free, close-up view [1]; feedback helps capture [2]
- Reading a mark is NOT verifying gold purity, authenticity or BIS certification
- Design target: transcription RELIABILITY, not only accuracy

Problem Statement

- Input: frame x with a hallmark-style inscription y (unknown to the system)
- Output: transcription $\hat{y}$, calibrated score p(x), decision
  - ACCEPT if $p(x) \geq \tau$ | RETAKE + hint | NO MARK
- Correct: $c = 1[\hat{y} = y]$ (whitespace-insensitive exact match)
- Goal: keep accepted-error rate below a target at useful coverage
- Out of scope: authenticity, purity, BIS certification, hallmark-standard compliance

Objectives

- O1 Reproducible, labelled SYNTHETIC benchmark, item-level split
- O2 Benchmark OCR engines and preprocessing
- O3 Does quality + OCR-uncertainty beat raw OCR confidence?
- O4 Calibrated, risk-targeted acceptance; measure threshold transfer
- O5 Recapture guidance evaluated against known causes
- O6 Simulator that runs the REAL pipeline
- O7 Wrist unit <-> companion interface: one HTTP POST, receiver tested
- O8 Wrist electronics + camera/illumination design; firmware compiled and booted in Wokwi (no camera there)
- O9 Image acquisition: OpenCV webcam + image upload feeding the pipeline

Literature Review (verified sources only)

- Mark analysis: goldsmith's-mark retrieval in silverware [3]
- OCR: Tesseract [4], PP-OCR [5]; scene-text surveys [6][7]
- Confidence: over-confident networks [8], MSP baseline [9], scene-text calibration [10]
- Selective prediction: Chow [11], El-Yaniv [13], Geifman [12], survey [14]
- Image quality: no-reference gradient and contrast statistics [15][16]
- No verified peer-reviewed work on OCR of engraved hallmarks found

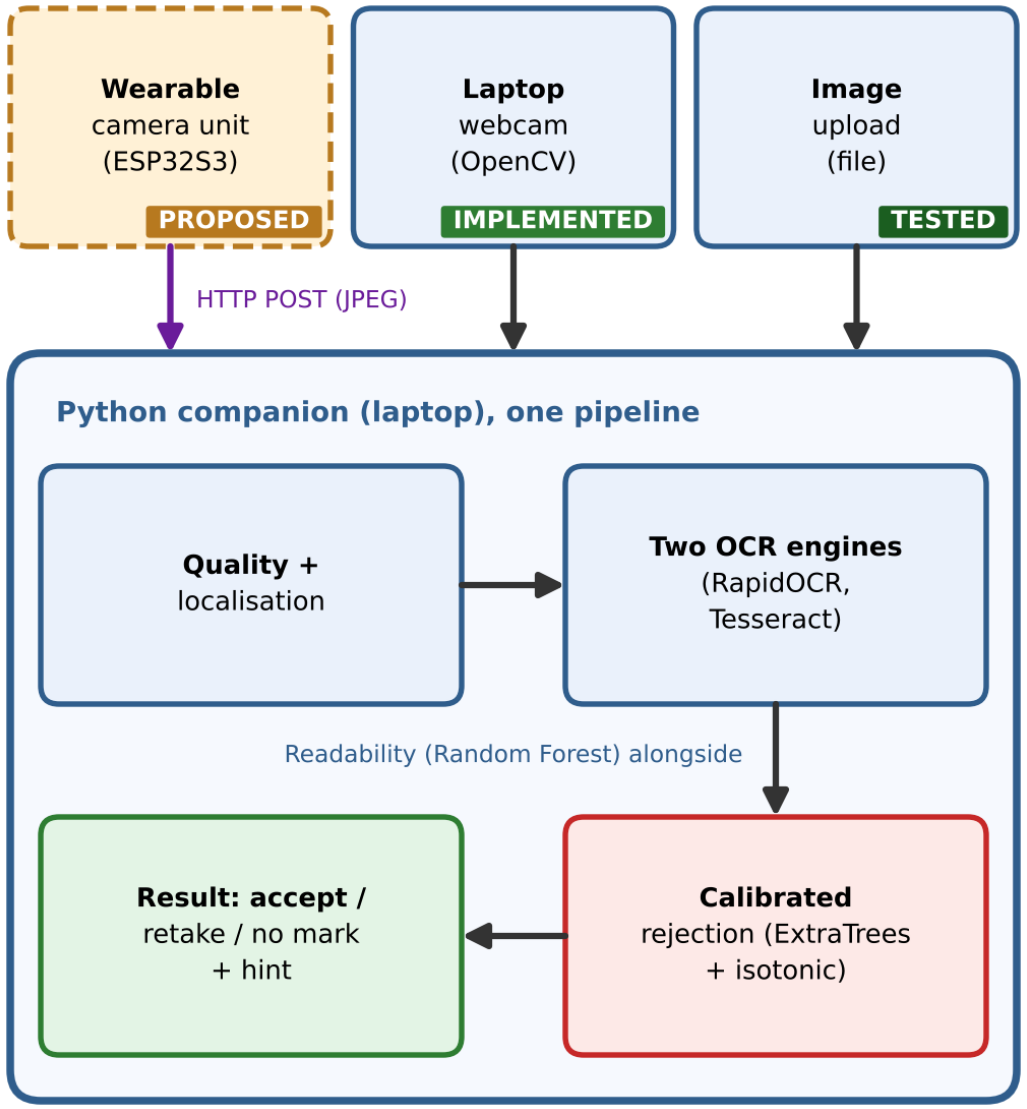
Research Gap and Contribution

- Gap: no verified work combining quality + OCR uncertainty + calibrated abstention + recapture advice for engraved text
- Contributions:
  - Synthetic generator with per-factor labels
  - OCR benchmark + preprocessing ablation
  - Calibrated separated rejector with risk-controlled thresholds
  - Rule-based recapture guidance, evaluated on known causes
  - Simulator running the real pipeline
  - Acquisition app: webcam (OpenCV) + upload -> same pipeline
  - Wrist camera + companion design, one-request HTTP interface, Wokwi firmware check
- Novelty not claimed beyond this combination

Wearable Concept: Wrist Camera + Companion

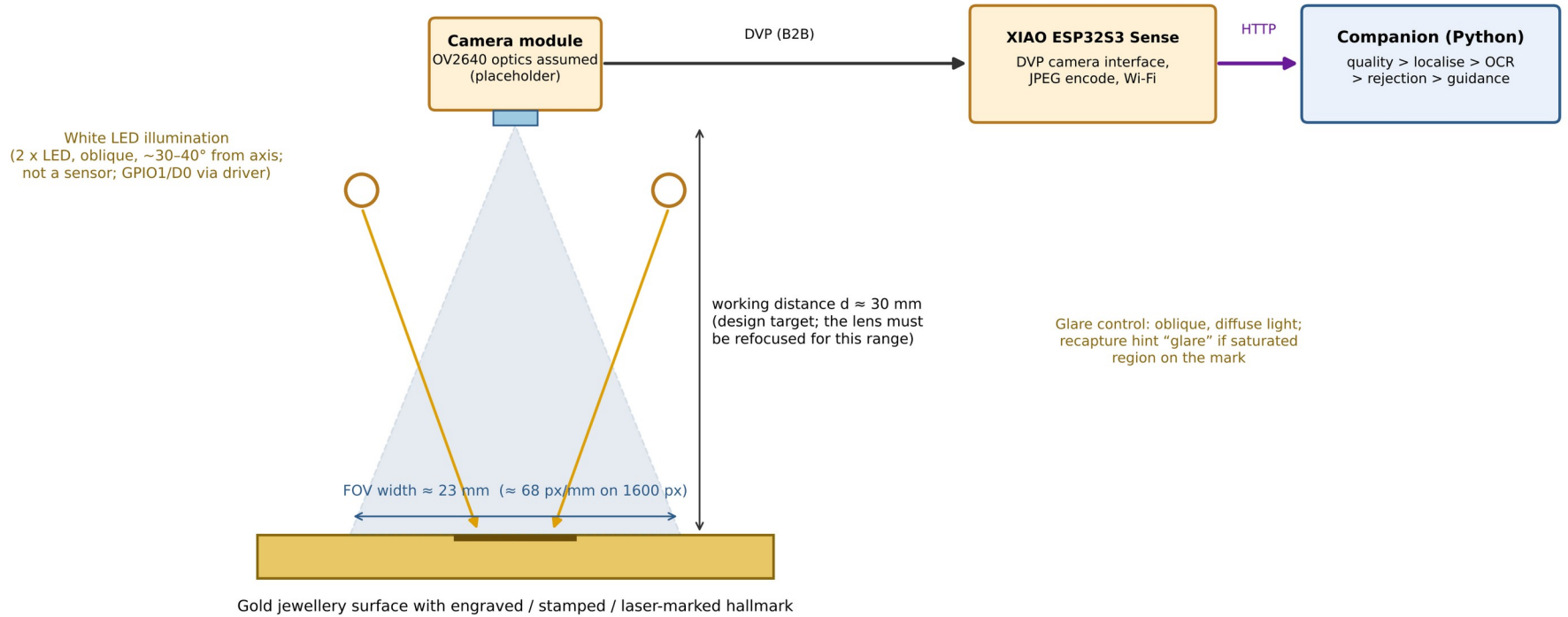
- Wrist unit: XIAO ESP32S3 Sense + proposed camera module (OV5640 AF, unconfirmed; primary sensor), white LED (illumination only), button, OLED
- Companion laptop: Python pipeline does ALL recognition (quality, localisation, 2 OCR engines, readability, rejection, guidance)
- Link: one HTTP POST of the JPEG over Wi-Fi; 3-line text reply shown on the OLED
- Status labels used throughout:
  - PROPOSED design: physical wrist unit, camera, LED driver (not built)
  - IMPLEMENTED: Python pipeline + HTTP receiver (synthetic images)
  - SIMULATED: firmware compile + boot in Wokwi (NO camera part exists)
  - TESTED: HTTP receiver with a Python client only

System Architecture



- Camera = primary sensor; LED = illumination only
- Recognition runs only on the companion laptop
- Wearable: one HTTP POST, 3-line text reply
- Wrist unit = PROPOSED, not built

Camera and Illumination Design



Example: a 1.5 mm character is ≈ 103 px tall at d = 30 mm; the synthetic images used in the experiments are 640x400 px, i.e. a lower resolution than this design.

- Proposed: XIAO ESP32S3 Sense + OV5640 AF; vendor says OV5640 or OV3660 may ship, so UNCONFIRMED [17]
- FOV and focus range: not listed by vendor
- Placeholder estimate from OV2640 [19], NOT OV5640 values: ~23 mm FOV, ~68 px/mm at 30 mm
- Illumination LED: D0/GPIO1 via NPN stage (not the sensor)
- Fallback: USB macro camera on the laptop
- NOT simulated in Wokwi; NOT built

Wearable Camera: Architecture Decision

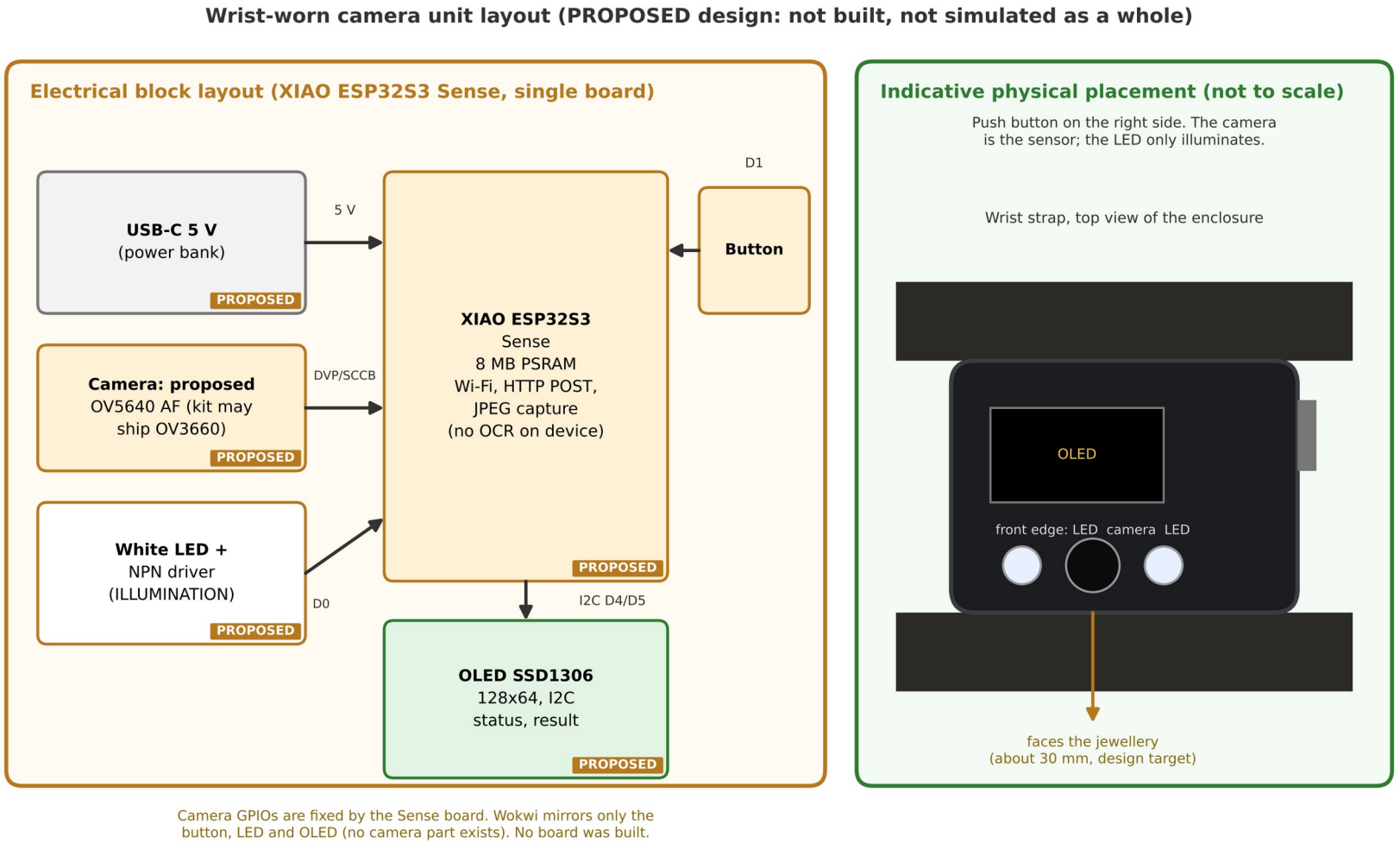
Part	Connection	Status
Camera (proposed OV5640 AF, unconfirmed)	board-to-board connector; no external wiring	NOT TESTED
SSD1306 OLED	D4 = SDA (GPIO5), D5 = SCL (GPIO6), 3V3, GND	boot only in Wokwi
CAPTURE button	D1 (GPIO2) to GND, internal pull-up	boot only in Wokwi
White LED stage	D0 (GPIO1) -> 1 k -> NPN base; LED + resistor from 5 V	PROPOSED
Power	USB-C 5 V (power bank); no battery design	PROPOSED
Link	Wi-Fi 2.4 GHz, one HTTP POST to /api/wearable/upload	receiver tested (Python client)

- Why not DevKit-C V4: 14 DVP signals, no camera connector, PSRAM unverified
- Why not ESP32-CAM: only 6 spare GPIOs, shared with SD/flash LED [18]
- Selected: XIAO ESP32S3 Sense, 8 MB PSRAM, single board [17]
- Whole unit is PROPOSED, NOT BUILT, NOT TESTED; the full wiring diagram is Fig. 2 of the paper

Wearable Camera: Verified vs Proposed

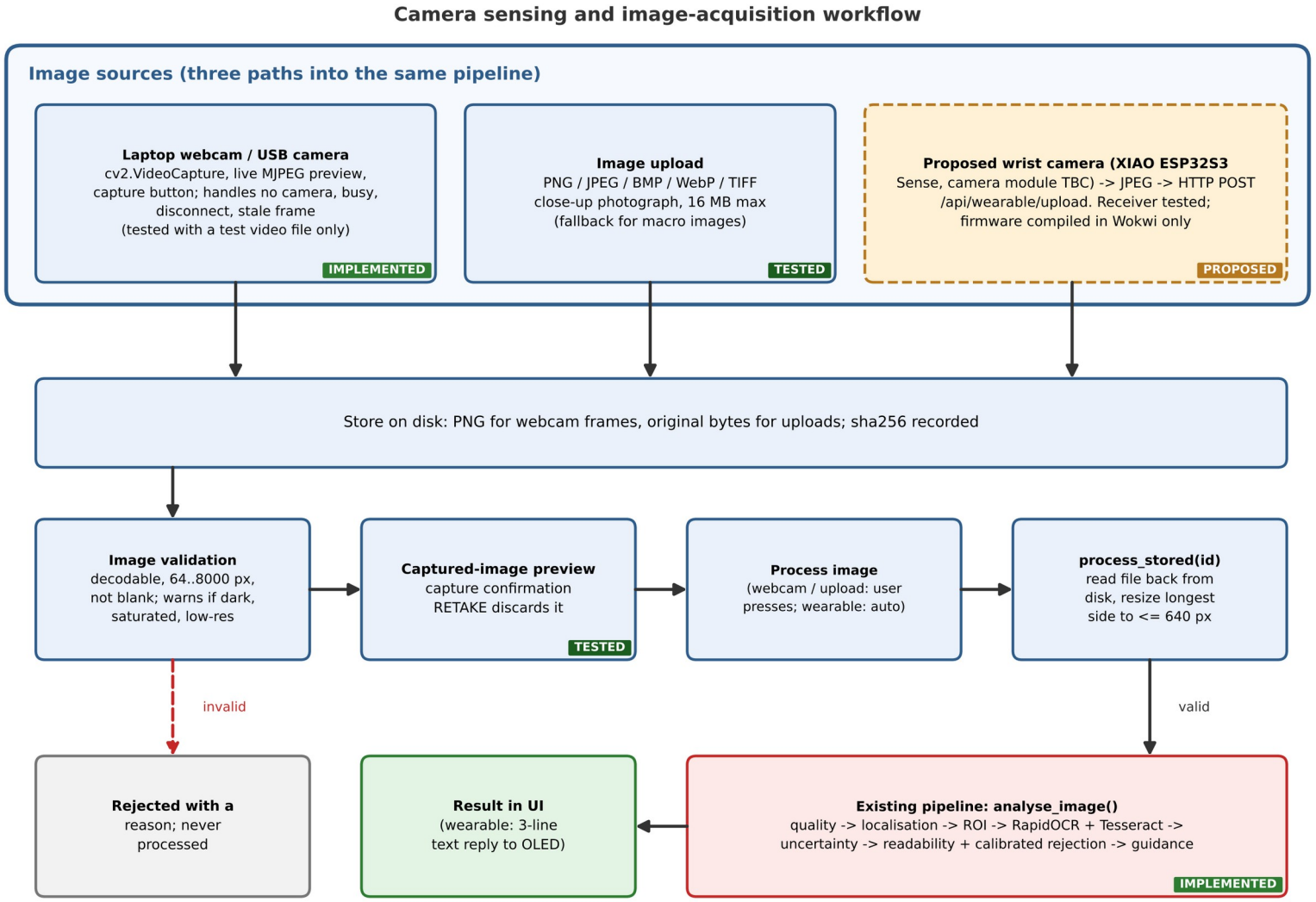
- DONE: board/camera/pin selection from vendor docs [17]
- SIMULATED: firmware compiles for XIAO ESP32-S3 and boots in Wokwi
- EXPECTED FAILURE: camera probe 0x106 (Wokwi has no camera part [20])
- TESTED (companion side): HTTP receiver, Python client
- NOT DONE: hardware, camera frame, firmware POST, round trip, power, optics
- Laptop webcam = separate path; NOT evidence of wearable integration

Proposed Wrist Unit: Electronics Layout



- Camera = primary sensor (proposed OV5640 AF module; unconfirmed)
- White LED = illumination only (NPN stage, 5 V)
- OLED D4/D5, button D1, LED D0
- Board runs capture + UI + comms; NO OCR
- PROPOSED: not built; Wokwi mirrors only button, LED, OLED

Image Acquisition Workflow (Implemented)



- Webcam (OpenCV): preview, capture, errors
- Upload: PNG/JPEG/BMP/WebP/TIFF
- Validate -> preview -> Retake / Process
- ONE entry point: process_stored()
- Webcam tested with a TEST VIDEO only (no camera in the sandbox)

Application Demo: Webcam Path

(a) Accepted reading (webcam path, test video)

3. PROCESSING AND RESULT

Done in 0.9 s.

ACCEPTED: READ OK | readability: readable

Transcription

79M9RA CEH 916

Calibrated probability the transcription is correct (p) vs acceptance threshold (tau)

p = 0.973 tau = 0.917 risk target = 0.05

Recapture guidance

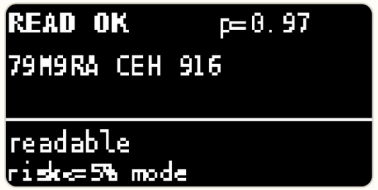
None needed: reading accepted at this risk target.



Input frame with detected hallmark region (green = accepted, red = rejected)



Extracted ROI passed to OCR



Emulated render of the wrist-unit OLED (not a photograph of hardware)

OCR engine	Input	Text	Confidence
primary: rapidocr	raw	79M9RA CEH 916	0.852
secondary: tesseract	red	79MSRA CEH 916	0.575

(b) Rejected capture with recapture guidance

3. PROCESSING AND RESULT

Done in 0.8 s.

RETAKE IMAGE | readability: partial

Transcription

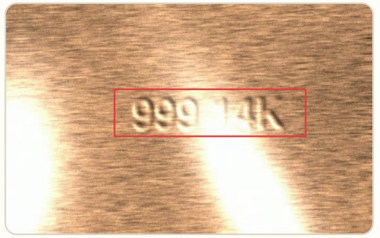
9991K

Calibrated probability the transcription is correct (p) vs acceptance threshold (tau)

p = 0.000 tau = 0.917 risk target = 0.05

Recapture guidance

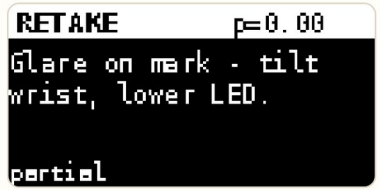
Glare on mark - tilt wrist, lower LED.



Input frame with detected hallmark region (green = accepted, red = rejected)



Extracted ROI passed to OCR



Emulated render of the wrist-unit OLED (not a photograph of hardware)

OCR engine	Input	Text	Confidence
primary: rapidocr	raw	9991K	0.618
secondary: tesseract	red	(empty)	0.000

Real screenshots (headless Chromium). Camera source = recorded TEST VIDEO of synthetic frames; no physical webcam. (a) accepted; (b) RETAKE.

Application Demo: Image Upload and Errors

3. PROCESSING AND RESULT

Done in 0.4 s.

ACCEPTED: READ OK | readability: readable

Transcription

24K YJNT5A ADC

Calibrated probability the transcription is correct (p) vs acceptance threshold (tau)

p = 0.978 tau = 0.917 risk target = 0.05

Recapture guidance

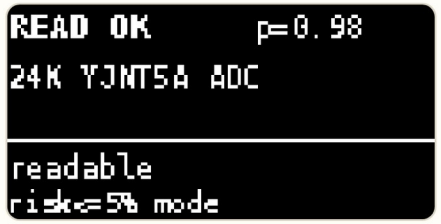
None needed: reading accepted at this risk target.



Input frame with detected hallmark region (green = accepted, red = rejected)

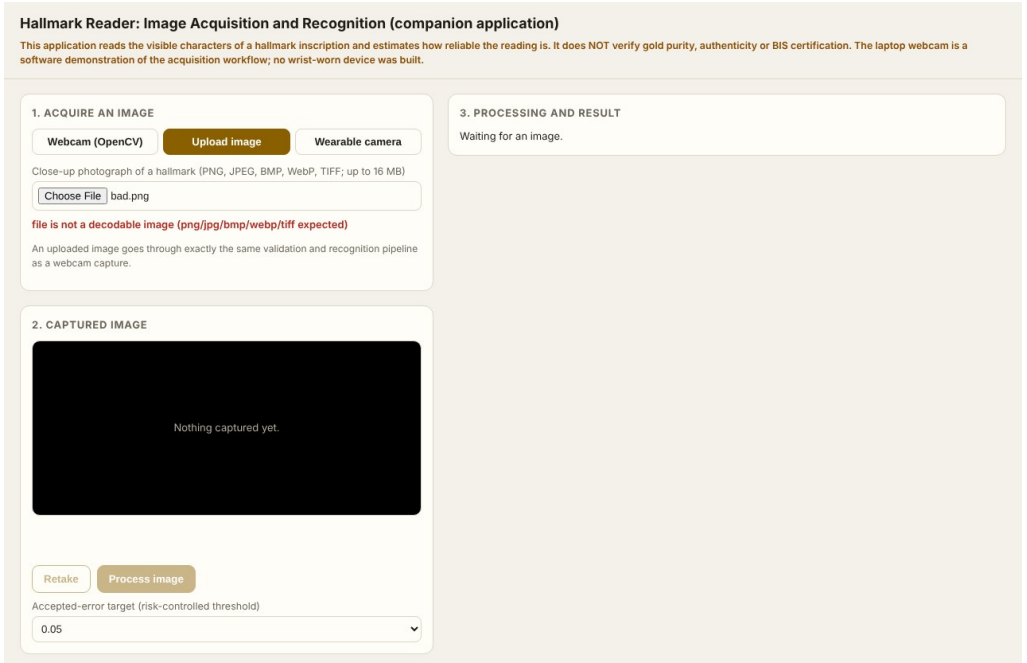


Extracted ROI passed to OCR



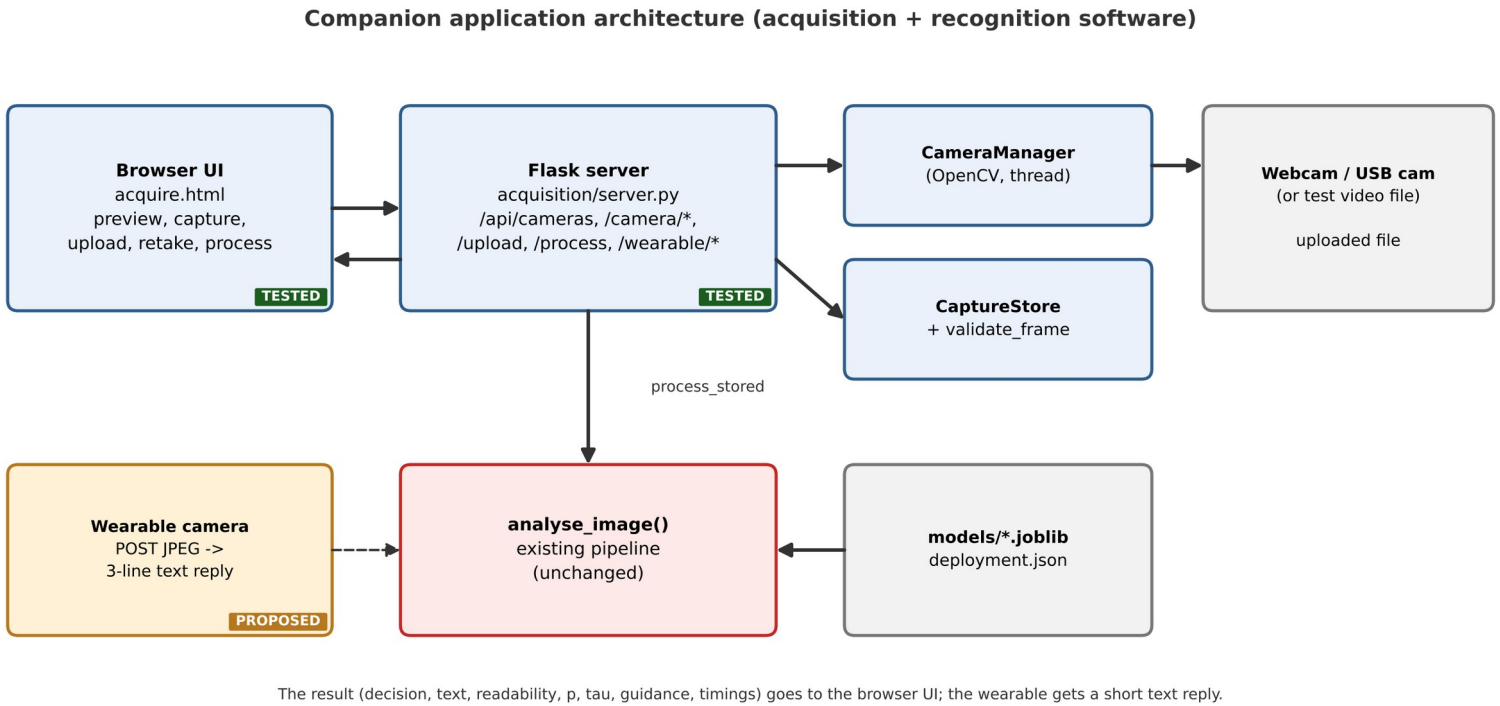
Emulated render of the wrist-unit OLED (not a photograph of hardware)

OCR engine	Input	Text	Confidence
primary: rapidocr	raw	24K YJNT5A ADC	0.828
secondary: tesseract	red	24K YJNT5AADC	0.345



Real screenshots. Left: upload result (SYNTHETIC test image, not a jewellery photograph). Right: corrupt upload refused.

Software Application and Tests

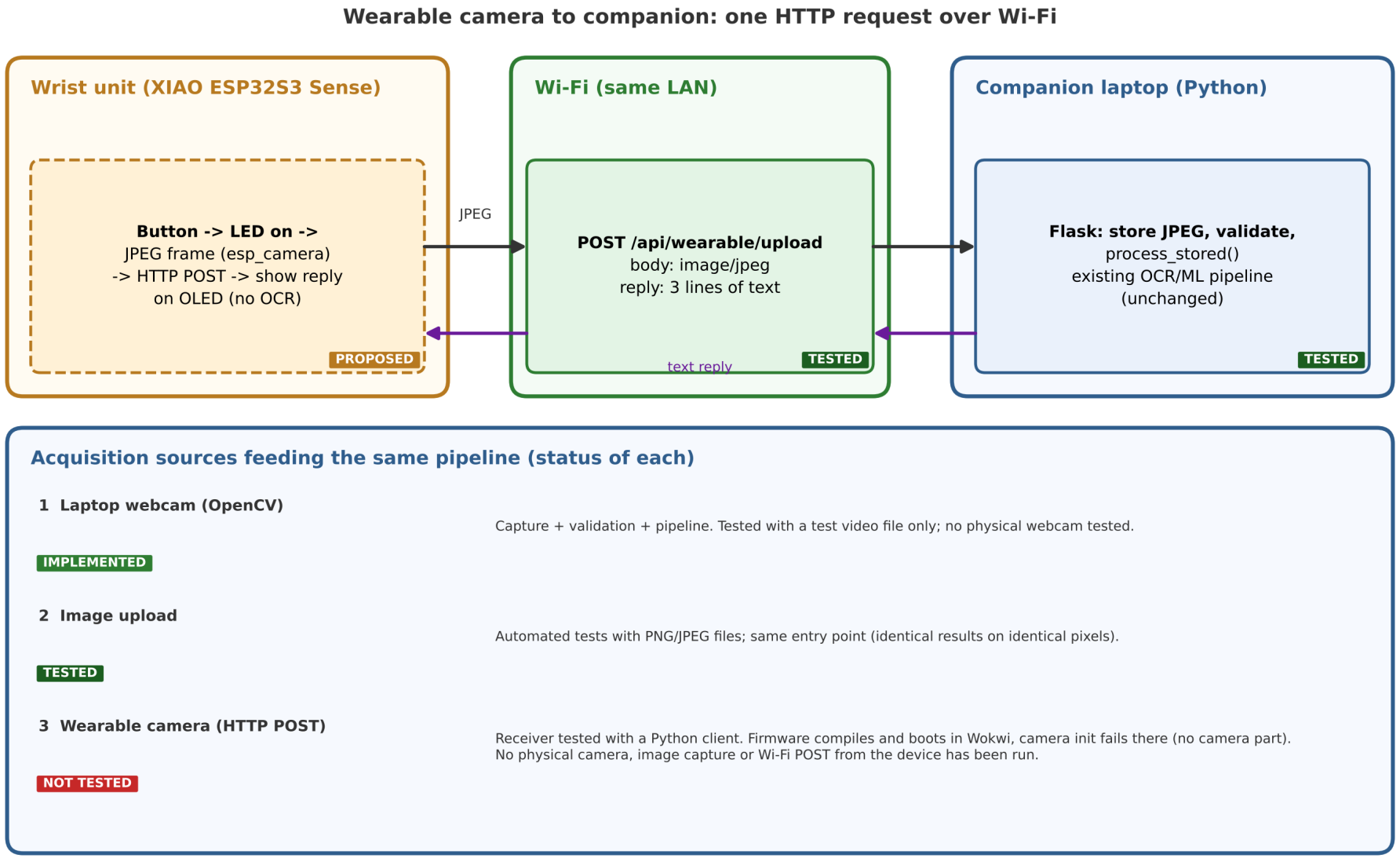


- 27 automated tests pass
- Webcam rows use a TEST VIDEO, not a camera
- Same pixels via webcam or upload -> identical result
- Corrupt / blank / tiny / unsupported files refused
- Disconnect detected; capture refused
- Every control clicked in headless Chromium

Wearable HTTP Interface

- POST /api/wearable/upload, body = JPEG, Content-Type image/jpeg
- Reply (text, 3 lines): READ OK | RETAKE | NO MARK
  - line 2: transcription (READ OK) or recapture hint
  - line 3: readability label
- Bad body: 400/422, first line BAD IMAGE; error: 500, first line ERROR
- Same process_stored() entry point as webcam and upload
- TESTED: Python client (4 automated tests); no hardware
- NOT tested: camera capture, firmware POST, Wi-Fi; plain HTTP, no TLS/authentication

Communication Architecture and Test Status

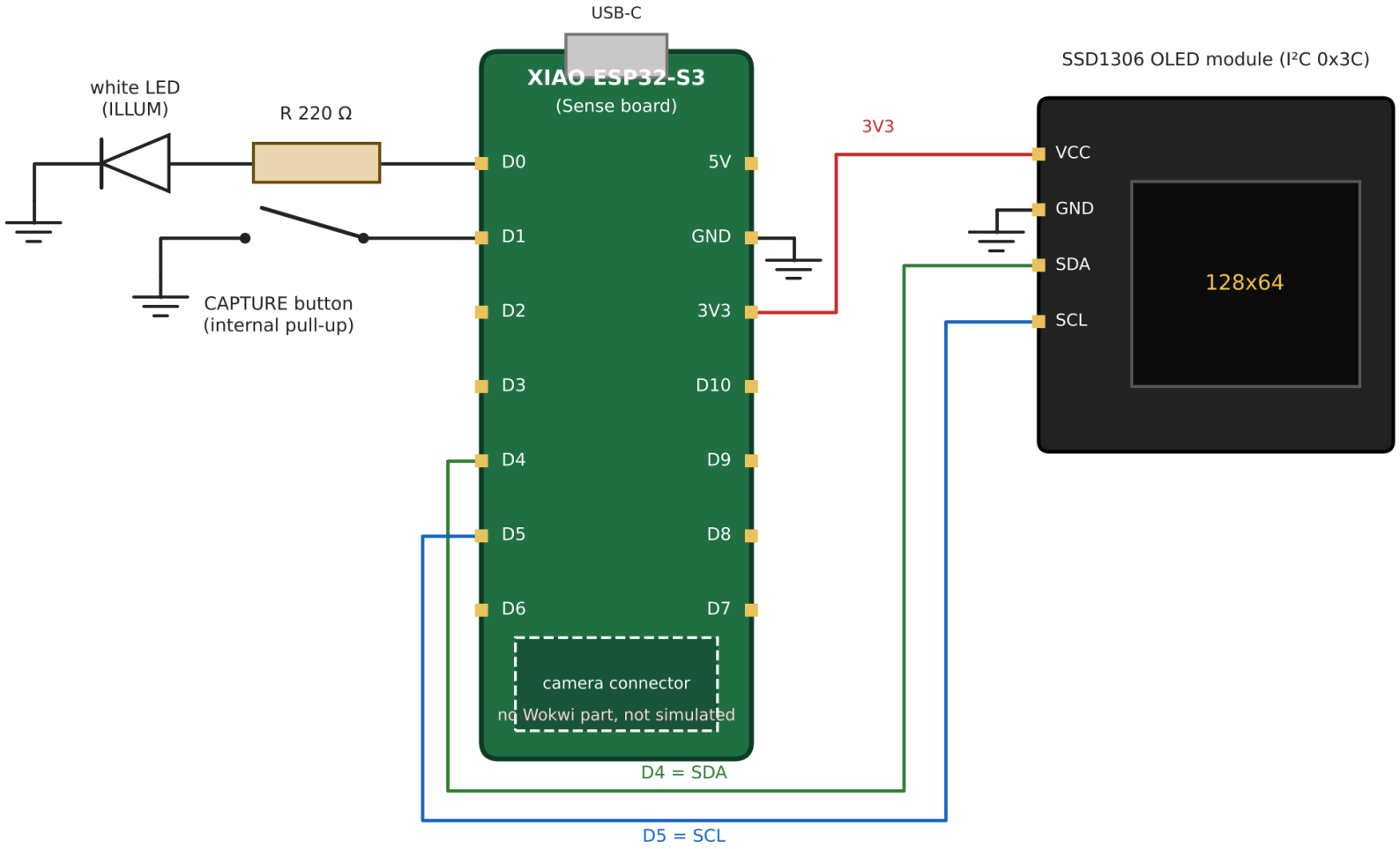


- One HTTP POST over Wi-Fi
- Webcam: implemented; test video only
- Upload: implemented and tested
- Wearable camera: NOT built/tested
- Receiver tested with Python client

Wokwi Project Circuit

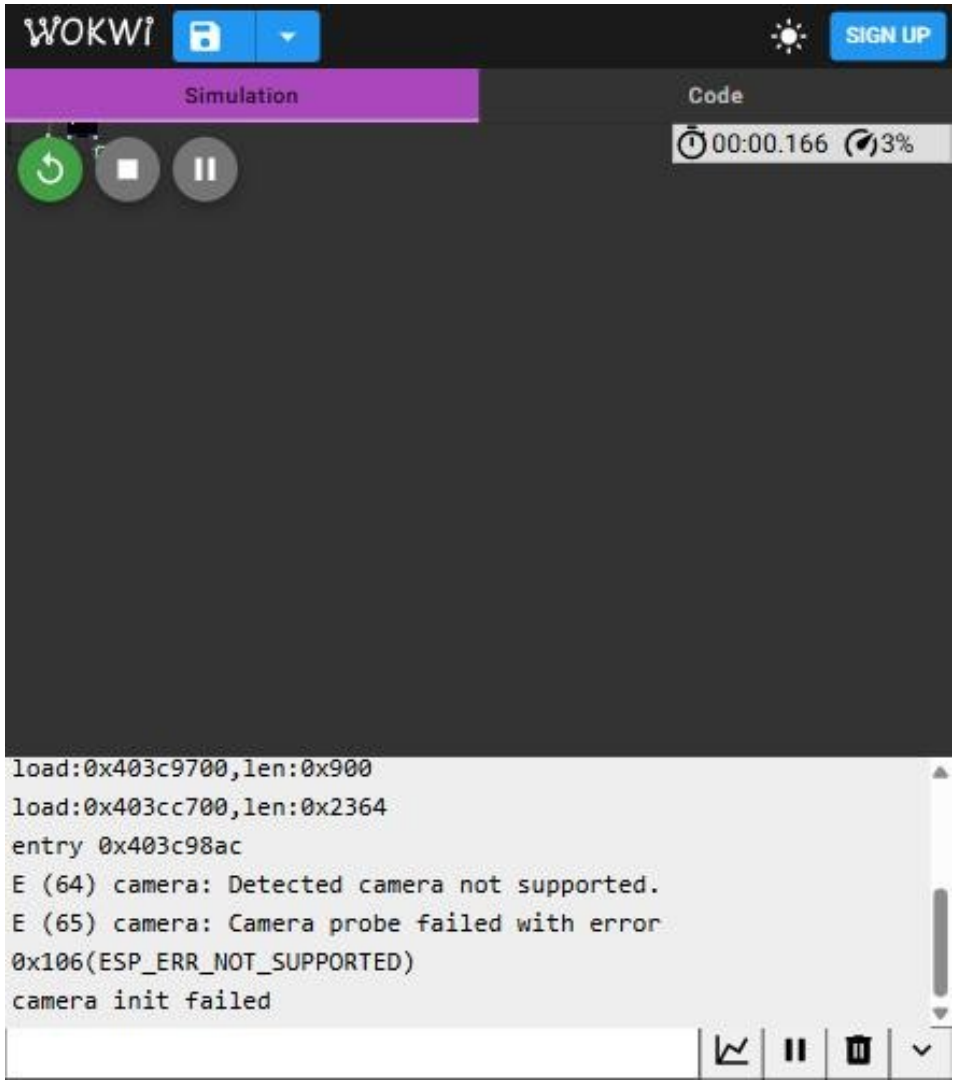
Wokwi project circuit (hardware/wokwi_xiao_s3/diagram.json): XIAO ESP32-S3, OLED, button, resistor, LED

Redrawn from diagram.json, not a Wokwi screenshot. Simplified: resistor-driven LED (the proposed unit uses a transistor stage). Wokwi has no camera part.



- Parts: XIAO ESP32-S3, SSD1306 OLED, button, 220 Ω resistor, white LED
- OLED: D4 = SDA, D5 = SCL, 3V3, GND
- Button: D1 to GND (internal pull-up)
- LED: D0 via 220 Ω (Wokwi has no NPN part)
- Redrawn from diagram.json; NOT a Wokwi screenshot
- No camera part in Wokwi: camera NOT simulated

Wokwi Firmware Check (No Camera)



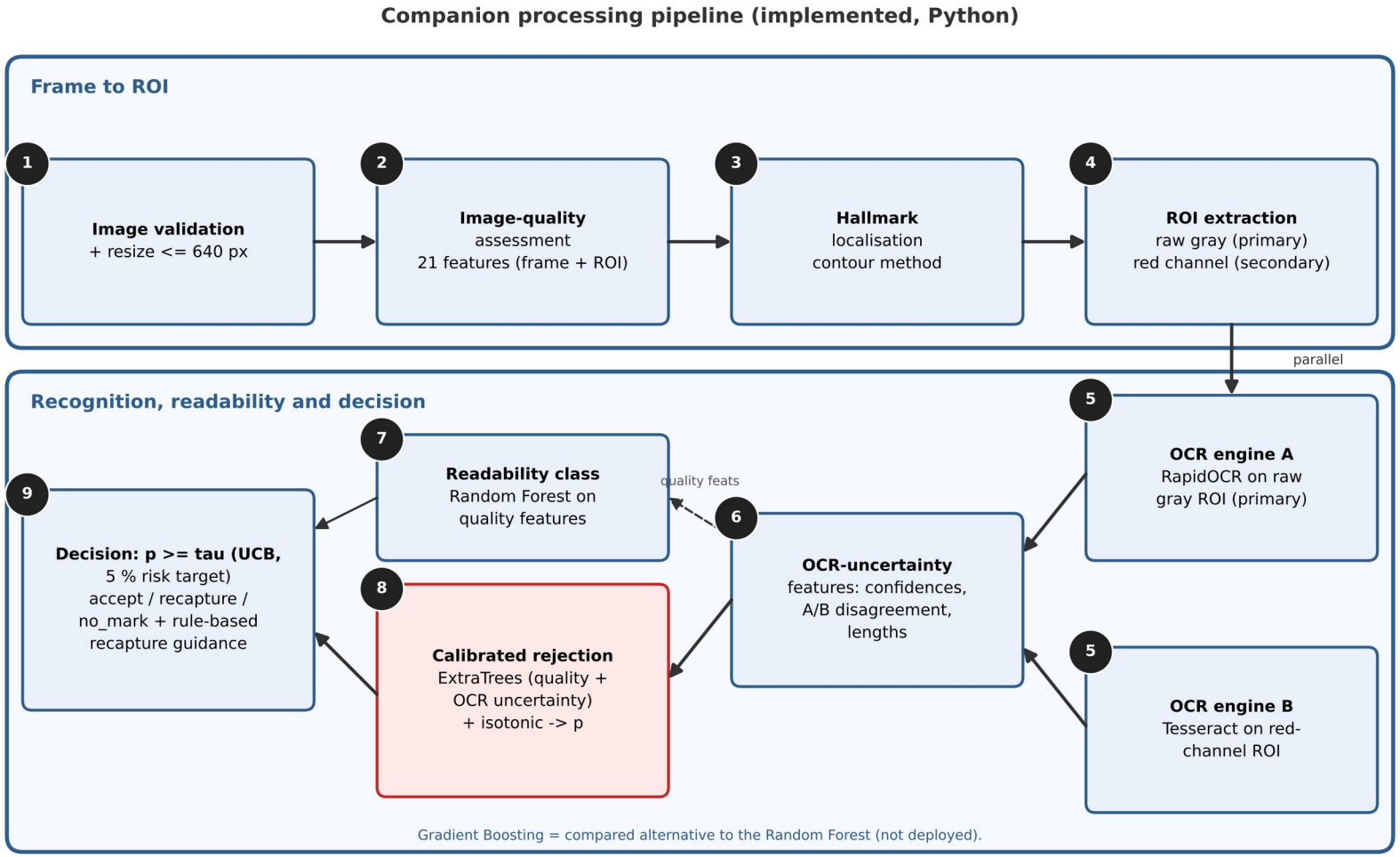
- Wokwi XIAO ESP32-S3 + SSD1306 OLED + button + LED
- Observed: compile, boot, camera probe fails (ESP_ERR_NOT_SUPPORTED)
- Expected: Wokwi has no camera part [20]
- NOT verified: button, OLED screens, Wi-Fi POST, any frame, round trip
- No claim that Wokwi tested the camera path

Dataset (100 % synthetic)



- 240 items × 6 acquisitions = 1440 images
- Split by ITEM: 144 / 48 / 48
- Designed to resemble gold-jewellery hallmark-style markings
- NOT real photographs; NOT official BIS samples
- Readability = generator-defined proxy
- Results = synthetic benchmark only

Companion Pipeline and Model Roles



- Primary OCR: RapidOCR (raw gray)
- Secondary OCR: Tesseract (red channel)
- Rejector: ExtraTrees + isotonic calibration
- Readability: Random Forest on quality features
- Gradient Boosting: compared, NOT deployed
- Guidance: rule-based

OCR and ML Workflow

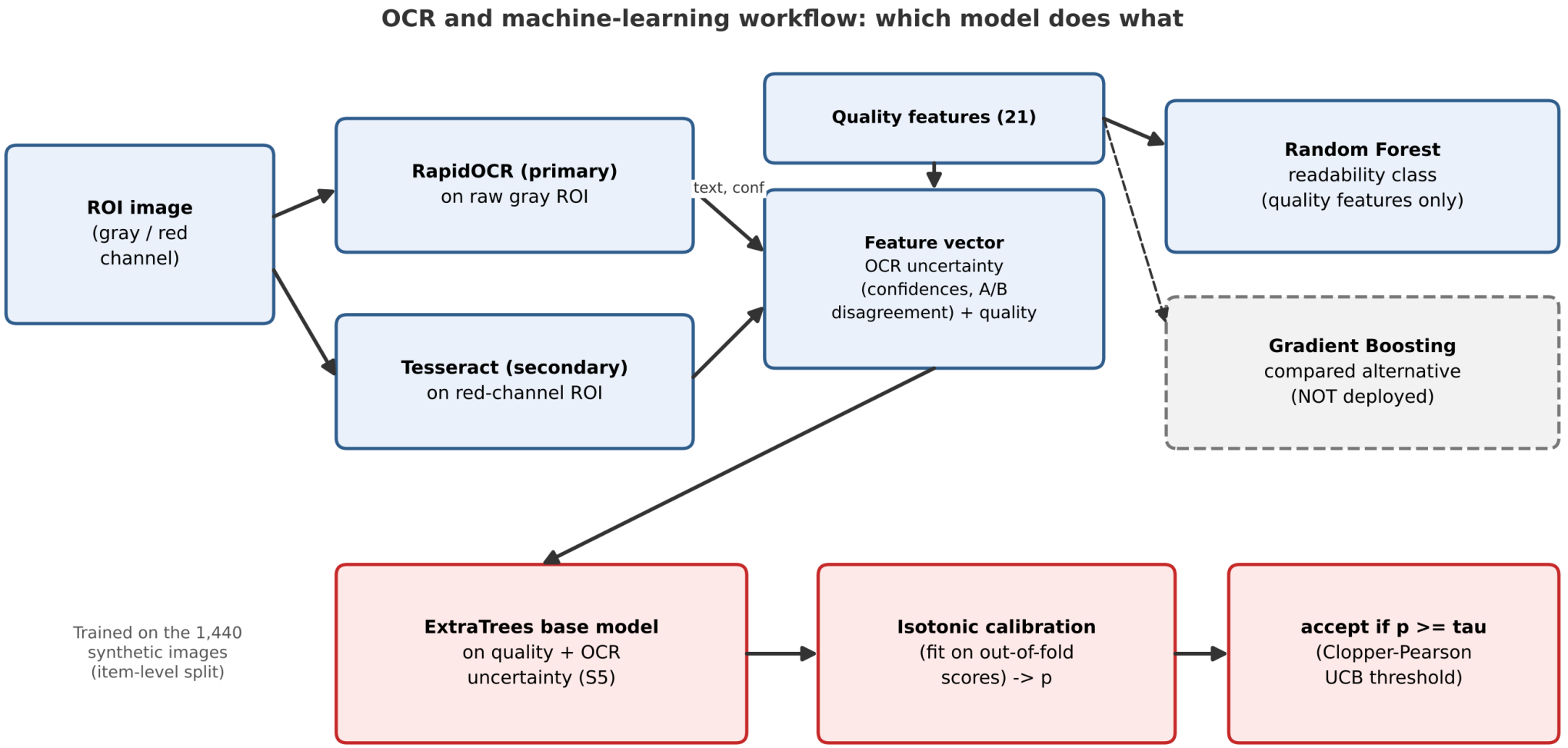
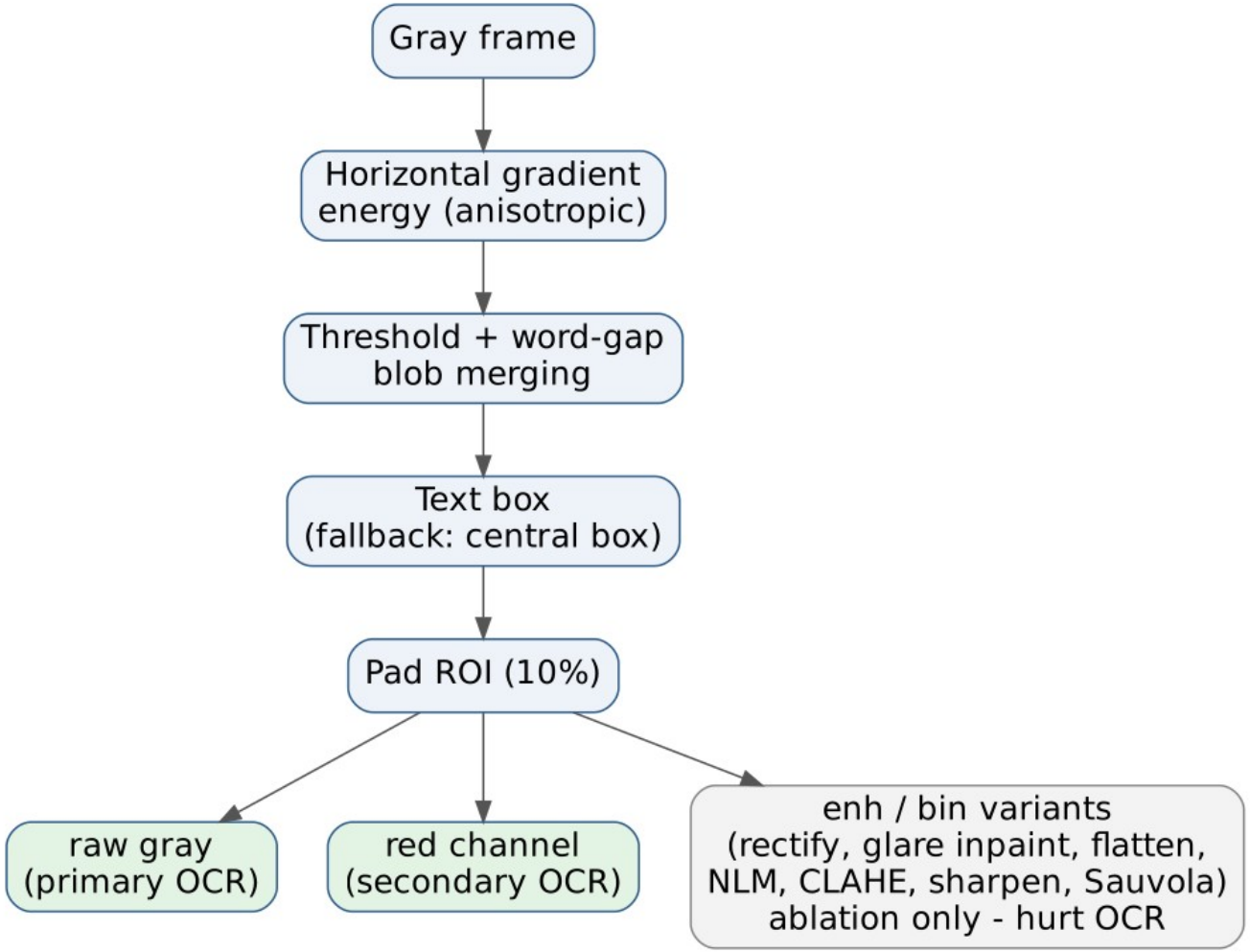


Image-Processing Pipeline

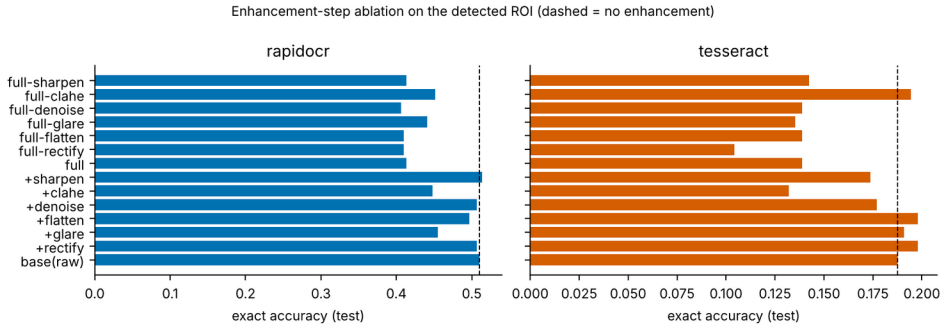


method	iou ge 0 5	latency ms median
contour	0.757	6.154
mser	0.656	22.521
ocrdet	0.809	133.041
full	0.000	0.001

Localisation (test). full = full-frame baseline

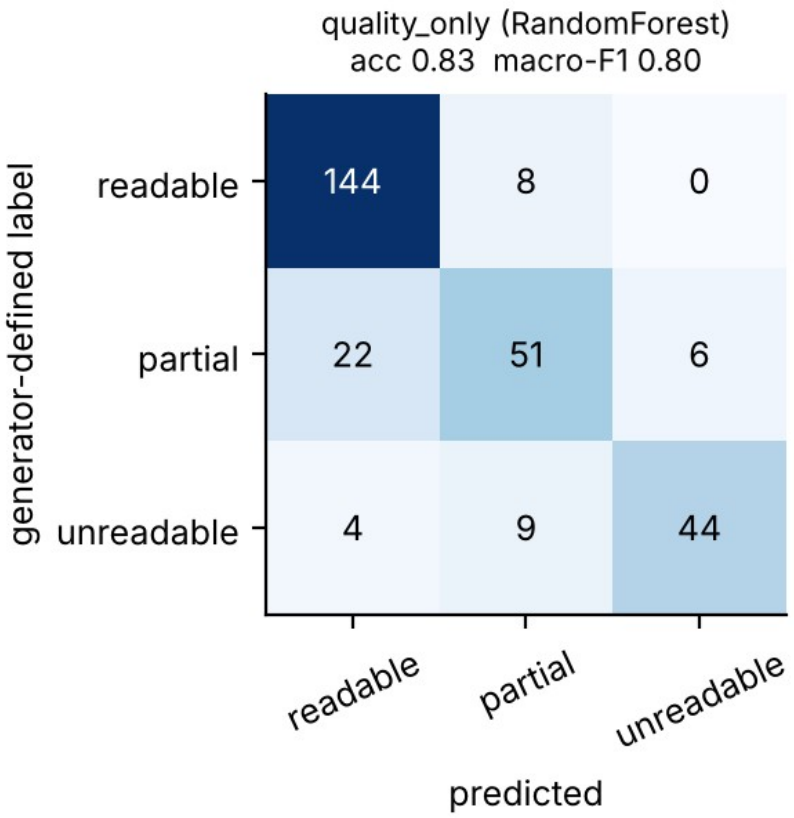
OCR Methodology

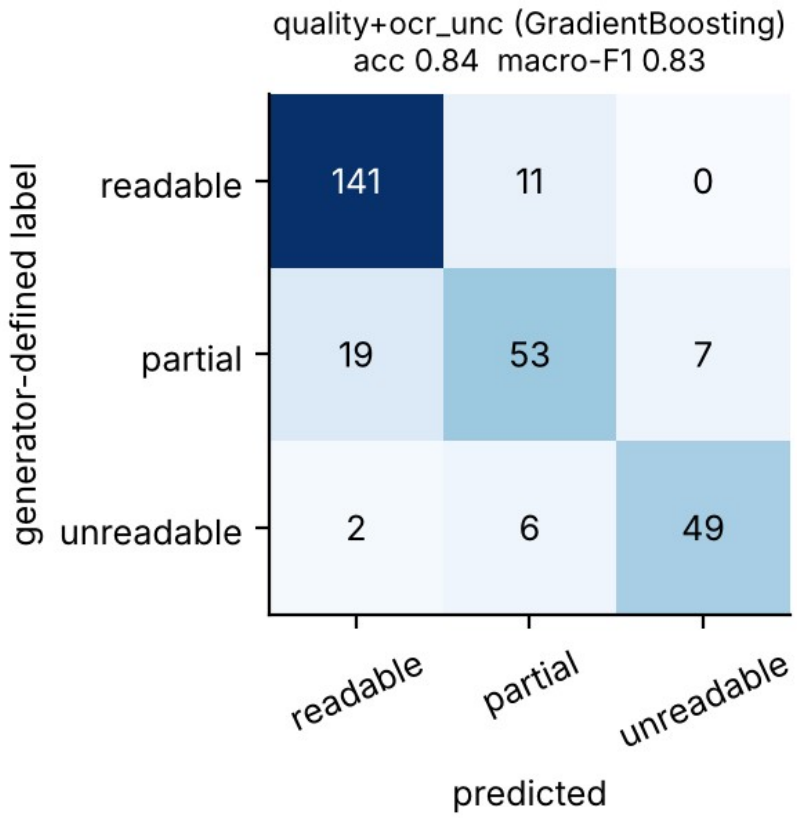
engine	input	test CER	test exact
tesseract	detected ROI, raw	0.632	0.188
tesseract	detected ROI, red	0.579	0.205
tesseract	detected ROI, enh	0.714	0.139
tesseract	detected ROI, bin	0.783	0.083
tesseract	GT-box ROI, raw	0.558	0.222
tesseract	GT-box ROI, enh	0.633	0.156
rapidocr	detected ROI, raw	0.271	0.510
rapidocr	detected ROI, red	0.286	0.472
rapidocr	detected ROI, enh	0.342	0.413
rapidocr	detected ROI, bin	0.469	0.205
rapidocr	GT-box ROI, raw	0.197	0.580
rapidocr	GT-box ROI, enh	0.261	0.431



Ablation: enhancement stages did not help these engines (negative result, synthetic data).

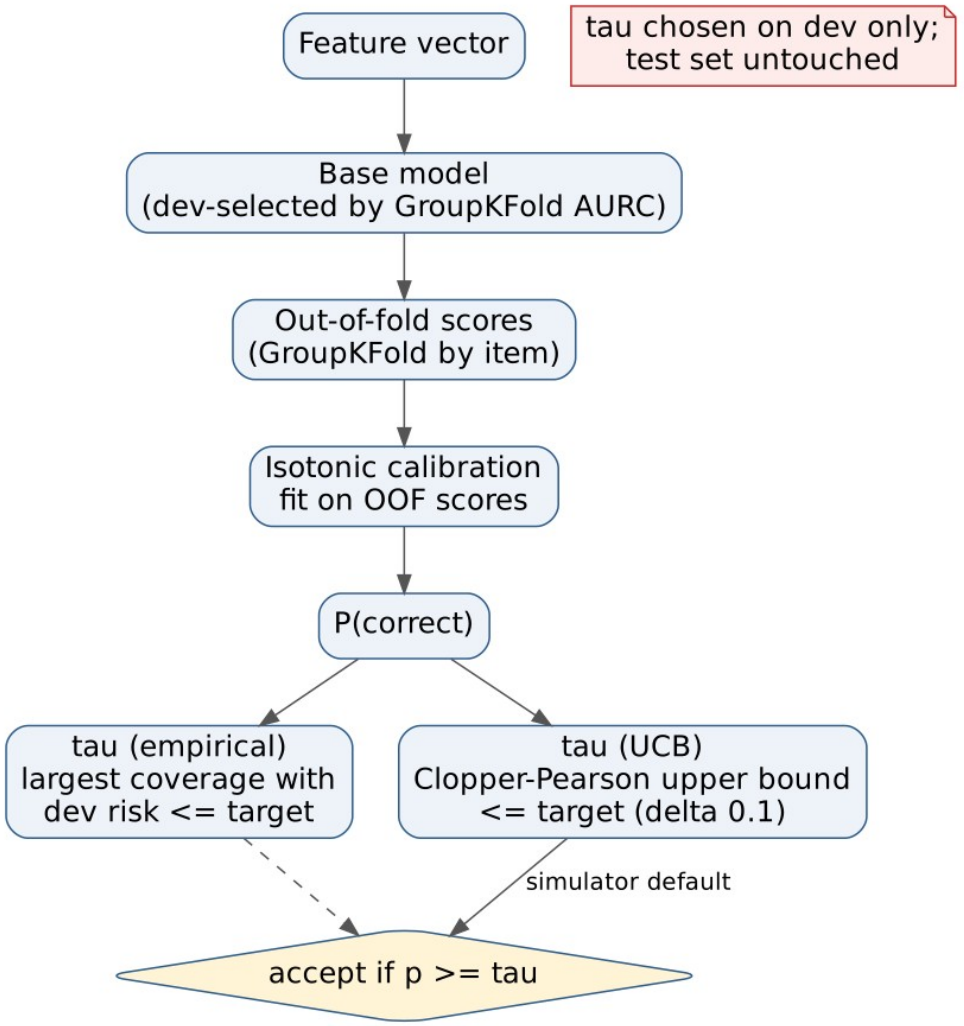
Readability Classification

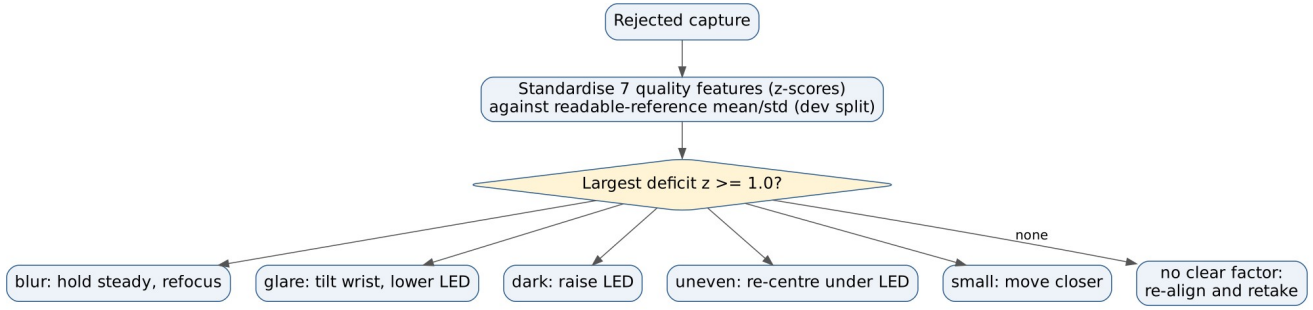




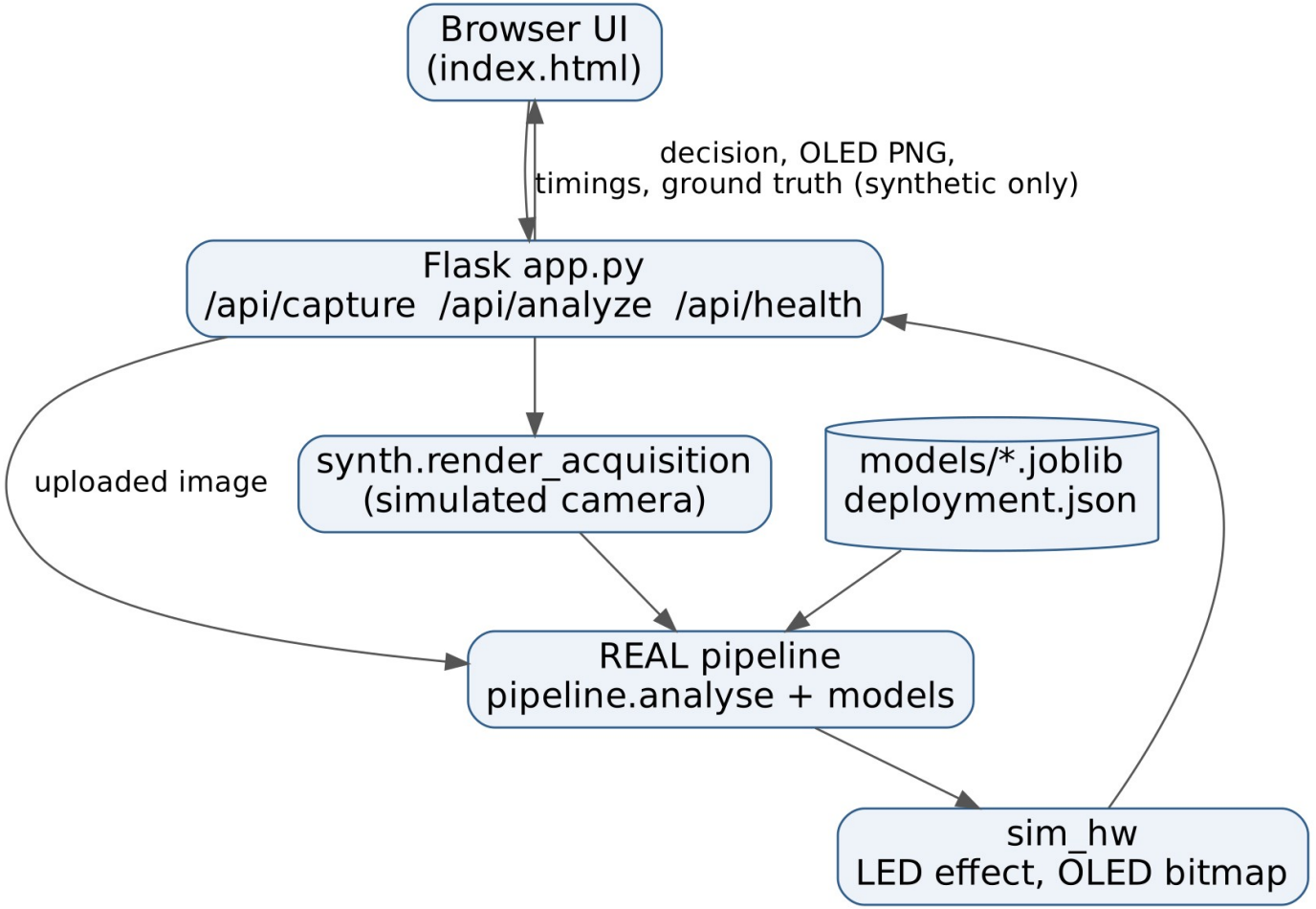
Test confusion matrices (rows: generator proxy label). Quality-only left; quality + OCR-uncertainty right.

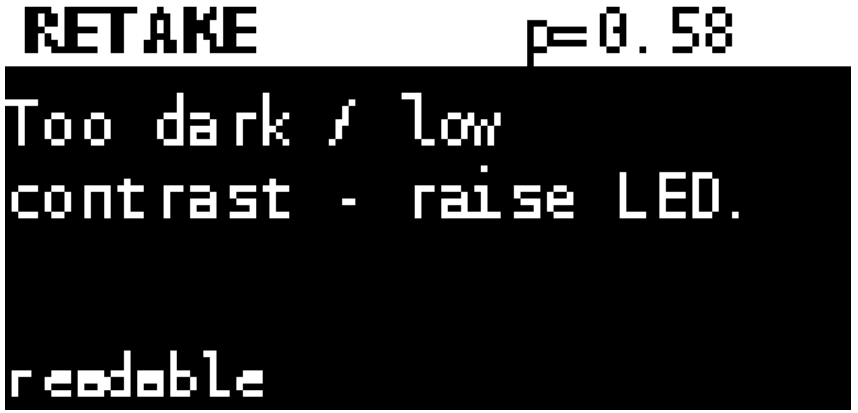
Confidence-Based Rejection and Recapture





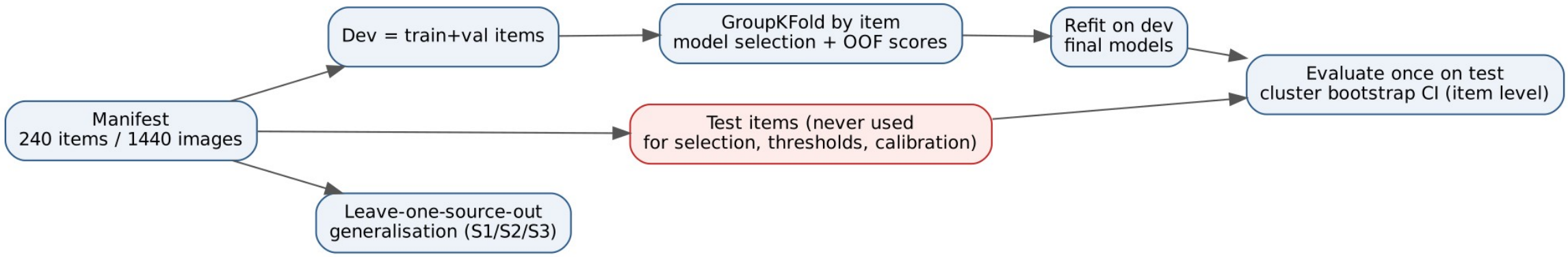
Original Simulator (Synthetic Camera)



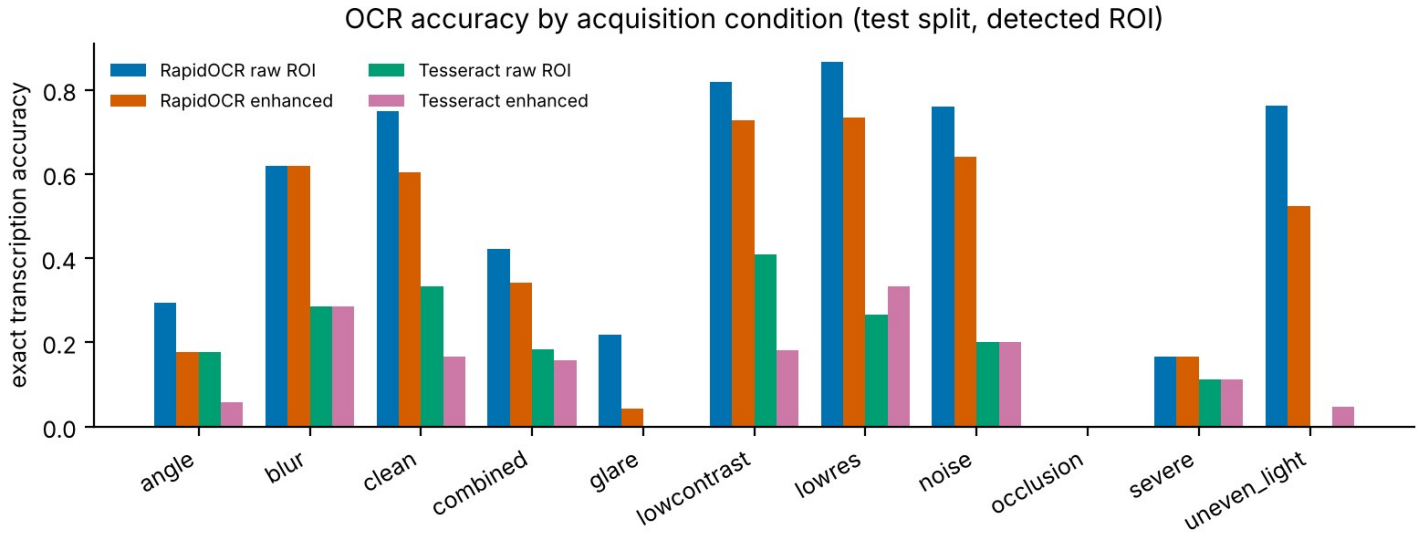


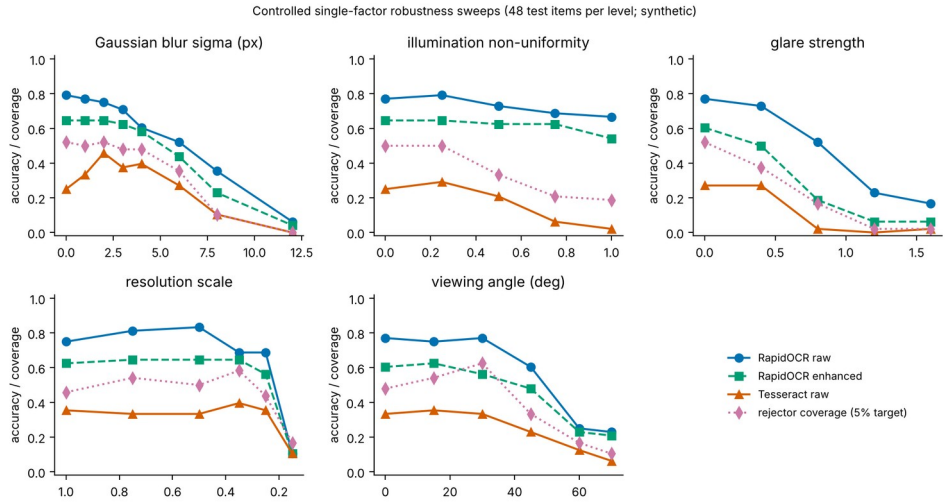
- Simulated camera = synthetic renderer (not webcam)
- Real pipeline + saved models
- 27 automated tests pass overall

Experimental Methodology

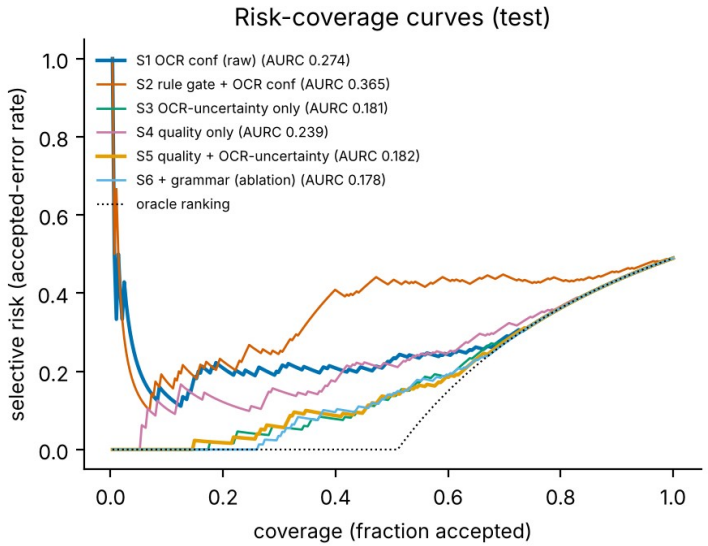


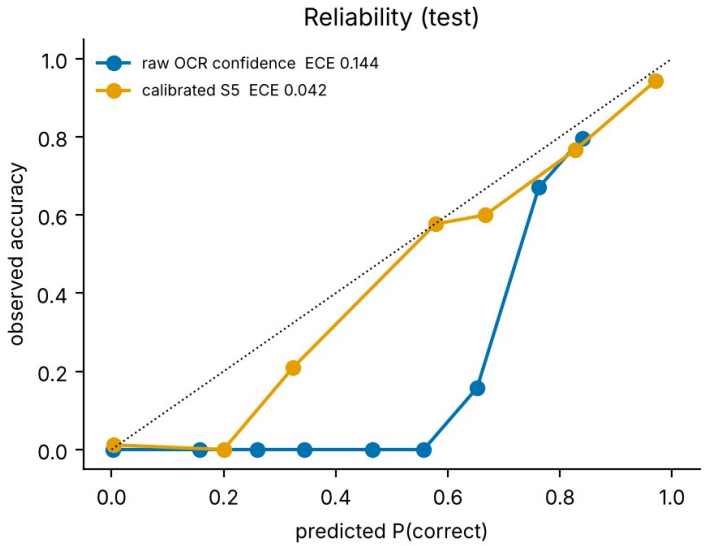
Results: OCR by Condition





Results: Rejection





comparison	delta AURC	CI95
S5 vs S3	0.000	[-0.009, +0.010]
S4 vs S1	-0.034	[-0.084, +0.009]
S3 vs S1	-0.090	[-0.146, -0.041]
S5 vs S4	-0.056	[-0.091, -0.028]

Paired ΔAURC (negative favours the first).

Results: Thresholds, Guidance, System

- S5 @5 % target: 87 accepted, 5 errors (5.7 %); coverage 0.30
- Dev thresholds do not transfer exactly; 48 test items only
- Guidance top-1 hit 0.674 vs random 0.185 (n=135)
- Leave-one-source-out: learned rejector beats raw confidence on all 3 sources
- Latency (sandbox CPU): median 264 ms, p95 340 ms
- Peak RSS 176 MB (not measured on Pi)

Discussion

- Learned OCR-uncertainty >> raw OCR confidence
- Quality features: no extra AURC gain here; they drive guidance
- Enhancement hurt RapidOCR (negative result)
- Threshold transfer is the practical weak point

Limitations

- Synthetic data only: no real gold jewellery imaged
- Labels defined from generator parameters
- Small test split (48 items): wide CIs
- Off-the-shelf OCR, no fine-tuning
- No hardware built; wearable camera never run; camera and transistor not simulable in Wokwi
- Device <-> companion link never run; HTTP receiver tested only with a Python client
- No user study; no Pi measurements
- Results do NOT establish real-jewellery performance

Conclusion and Future Work

- On the synthetic benchmark: calibrated OCR-uncertainty rejector beats raw confidence (ΔAURC −0.09, CI excl. 0)
- Quality features → explainable recapture hints; no extra AURC
- Design: proposed wrist camera unit + implemented Python companion + one-request HTTP receiver tested with a Python client
- Future: real annotated photographs, fine-tuned recogniser, build the wrist unit, send a real frame from the device, authenticated link, user study

References

[1] K. Ohnishi, A. Kanehira, A. Kanezaki, T. Harada, "Recognizing Activities of Daily Living with a Wrist-Mounted Camera," Proc. IEEE CVPR, 2016. [Online]. Available: https://openaccess.thecvf.com/content_cvpr_2016/html/Ohnishi_Recognizing_Activities_of_CVPR_2016_paper.html

[2] R. Shilkrot, J. Huber, M. E. Wong, P. Maes, S. Nanayakkara, "FingerReader: A Wearable Device to Explore Printed Text on the Go," Proc. ACM CHI 2015, Seoul, 2015. [Online]. Available: http://hdl.handle.net/1721.1/95971

[3] A. Tiwari, V. Christlein, M. Fichtner, F. Gohlke, B. Schübel, T. Witting, H. Zech, M. Zinnen, "Automated Goldsmith's Mark Retrieval in Silverware," arXiv preprint arXiv:2609.20509, 2026. [Online]. Available: https://arxiv.org/abs/2609.20509

[4] R. Smith, "An Overview of the Tesseract OCR Engine," Proc. 9th Int. Conf. Document Analysis and Recognition (ICDAR), 2007. [Online]. Available: https://research.google/pubs/an-overview-of-the-tesseract-ocr-engine/

[5] Y. Du, C. Li, R. Guo, X. Yin, W. Liu, J. Zhou, Y. Bai, Z. Yu, Y. Yang, Q. Dang, H. Wang, "PP-OCR: A Practical Ultra Lightweight OCR System," arXiv preprint arXiv:2009.09941, 2020. [Online]. Available: https://arxiv.org/abs/2009.09941

[6] S. Long, X. He, C. Yao, "Scene Text Detection and Recognition: The Deep Learning Era," Int. J. Comput. Vis., vol. 129 (arXiv:1811.04256), 2021. [Online]. Available: https://arxiv.org/abs/1811.04256

[7] X. Chen, L. Jin, Y. Zhu, C. Luo, T. Wang, "Text Recognition in the Wild: A Survey," arXiv preprint arXiv:2005.03492, 2020. [Online]. Available: https://arxiv.org/abs/2005.03492

[8] C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, "On Calibration of Modern Neural Networks," Proc. ICML, PMLR vol. 70, 2017. [Online]. Available: https://proceedings.mlr.press/v70/guo17a.html

[9] D. Hendrycks, K. Gimpel, "A Baseline for Detecting Misclassified and Out-of-Distribution Examples in Neural Networks," Proc. ICLR, 2017. [Online]. Available: https://arxiv.org/abs/1610.02136

[10] R. Slossberg, O. Anschel, A. Markovitz, R. Litman, A. Aberdam, S. Tsiper, S. Mazor, J. Wu, R. Manmatha, "On Calibration of Scene-Text Recognition Models," Computer Vision - ECCV 2022 Workshops, pp. 263-279, 2022. [Online]. Available: https://doi.org/10.1007/978-3-031-25069-9_18

[11] C. K. Chow, "On optimum recognition error and reject tradeoff," IEEE Trans. Inf. Theory, vol. 16, no. 1, pp. 41-46, 1970. [Online]. Available: https://researchweb.draco.res.ibm.com/publications/on-optimum-recognition-error-and-reject-tradeoff

[12] Y. Geifman, R. El-Yaniv, "Selective Classification for Deep Neural Networks," Advances in Neural Information Processing Systems (NeurIPS) 30, 2017. [Online]. Available: https://arxiv.org/abs/1705.08500

[13] R. El-Yaniv, Y. Wiener, "On the Foundations of Noise-free Selective Classification," J. Mach. Learn. Res., vol. 11, 2010. [Online]. Available: https://jmlr.org/papers/v11/el-yaniv10a.html

[14] K. Hendrickx, L. Perini, D. Van der Plas, W. Meert, J. Davis, "Machine Learning with a Reject Option: A survey," arXiv preprint arXiv:2107.11277, 2021. [Online]. Available: https://arxiv.org/abs/2107.11277

[15] A. Mittal, A. K. Moorthy, A. C. Bovik, "No-Reference Image Quality Assessment in the Spatial Domain," IEEE Trans. Image Process., vol. 21, no. 12, pp. 4695-4708, 2012. [Online]. Available: https://doi.org/10.1109/TIP.2012.2214050

[16] H. Li, F. Zhu, J. Qiu, "Towards Document Image Quality Assessment: A Text Line Based Framework and A Synthetic Text Line Image Dataset," arXiv preprint arXiv:1906.01907, 2019. [Online]. Available: https://arxiv.org/abs/1906.01907

[17] Seeed Studio, "XIAO ESP32S3 (Sense) pin multiplexing and getting-started wiki pages," Seeed Studio wiki (manufacturer documentation), 2026. [Online]. Available: https://wiki.seeedstudio.com/xiao_esp32s3_pin_multiplexing/

[18] JLCPCB, "What Is the ESP32-CAM, and How to Wire, Power, and Program It?," JLCPCB blog (secondary source), 2026. [Online]. Available: https://jlcpcb.com/blog/esp32-cam-guide

[19] Arducam, "Arducam OV2640 Camera Module (2MP Mini CCM, M0031) product page," Arducam (manufacturer page), 2026. [Online]. Available: https://www.arducam.com/product/arducam-ov2640-camera-module-2mp-mini-ccm-compact-camera-modules-compatible-with-arduino_m0031esp32-esp8266-development-board-with-dvp-24-pin-interface_/

[20] Wokwi, "Wokwi supported hardware," Wokwi documentation, 2026. [Online]. Available: https://docs.wokwi.com/getting-started/supported-hardware